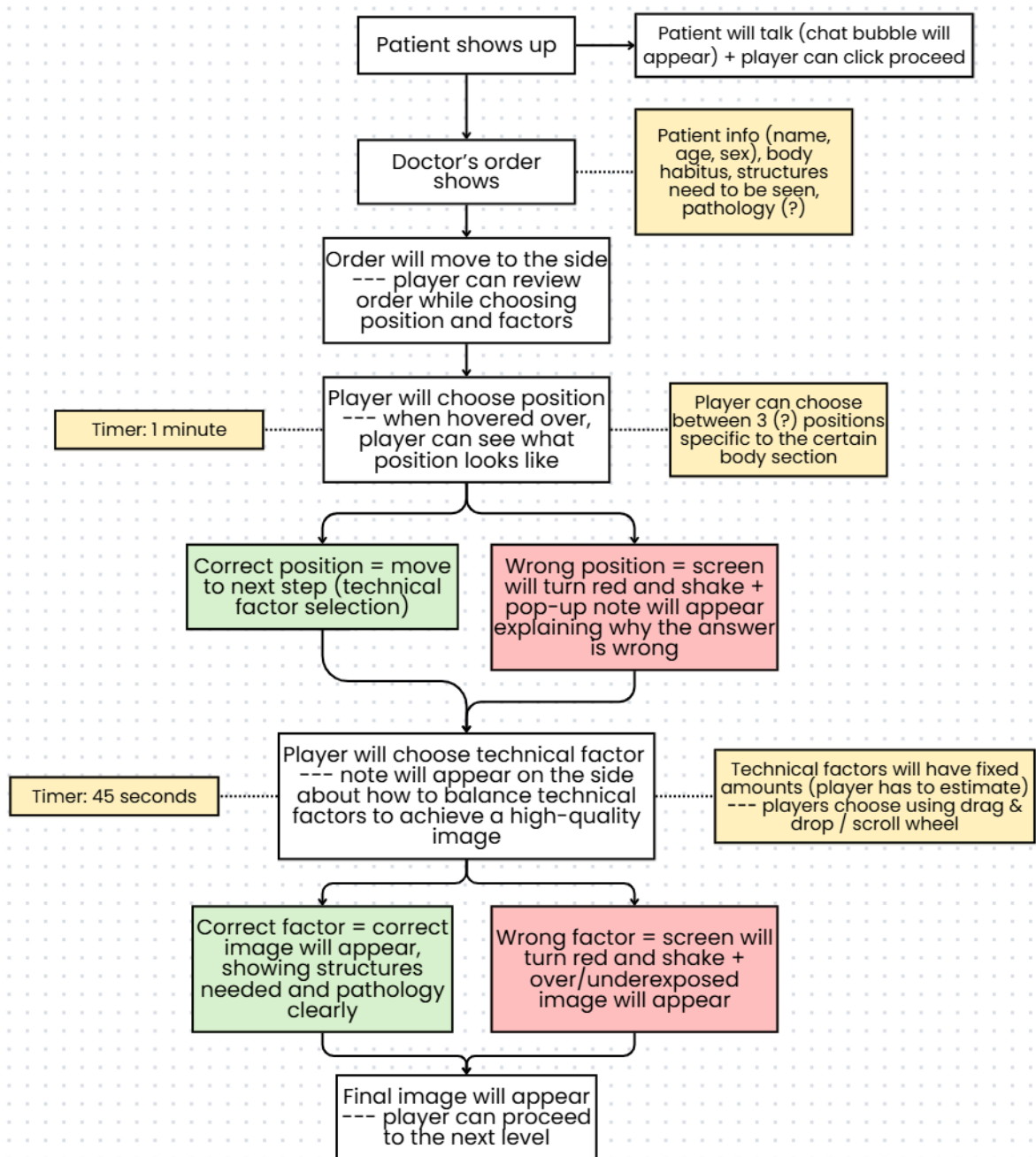


A. WEBSITE FLOW



B. DETAILS

Web- Based Game

Game Design Document

1. Game Overview

This Web based game is a level-based simulation game that puts the player in the role of a radiologic technologist (radtech). In each level, a patient arrives with a specific injury, and the player must guide them through the real-world X-ray workflow: reading the case file,

assessing the injury, positioning the patient, setting exposure factors (kVp and mAs), adjusting collimation, and finally capturing the exposure at the right moment. The game is designed to teach the logic and sequence of radiographic procedures through hands-on, mini-game-style interactions rather than passive reading.

The five levels progress through different regions of the body, each with its own positioning rules, exposure ranges, and collimation requirements increasing in difficulty as the player advances.

Genre: Simulation / Educational / Puzzle

Perspective: Top-down / screen-based UI (clinic room view transitioning into procedure-specific mini-screens)

Core Fantasy: Being a skilled Radtech who correctly diagnoses positioning, exposure, and collimation for real injuries.

2. Core Gameplay Loop

Every level follows the same seven-stage loop, with the specific details (injury, positioning, kVp/mAs range, collimation size) changing per level:

- Patient Intake - Patient enters and hands over their file.
- Read the Case File - Player opens the file to learn the exam type and injury.
- Assess the Patient - Player taps/zooms into the injured area, which highlights in red.
- Positioning - The player selects the correct positioning pose from a selection bar.
- Exposure Settings - Player dials in the correct kVp and mAs for the injury.
- Collimation - Player adjusts the X-ray beam field to match a preset marker size for the injury.
- Capture the Exposure - A quick-time event (QTE) where the player presses the exposure button at the right moment.

Successful completion of all seven stages ends the level with a success animation, unlocking the next (harder) level.

3. Title Screen & Menu

On opening the app, the player is greeted with a Title Screen containing three options:

- Start- Begins the game at the current/selected level
- Select Level- opens a level-select screen showing unlocked levels (1–5)
- Settings- adjusts game options (audio, difficulty, tutorial toggle, etc.)

4. Educational Intent

The overall design mirrors the actual clinical workflow of a radiologic technologist:

- Receive and interpret a patient's case/order.
- Visually and physically assess the injury.
- Position the patient correctly for the required view.
- Select appropriate technical exposure factors for the body part's density/sensitivity.
- Collimate to the smallest field necessary to capture the area of interest (a real radiation-safety principle: minimizing exposed area).

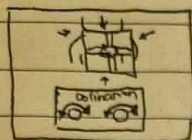
- Time the exposure correctly to avoid over/under-exposure.

By turning each of these into a distinct, scored interaction, the game reinforces correct sequencing and decision-making for radiography students in a low-stakes, replayable format.

C. REFERENCES/INSPIRATION



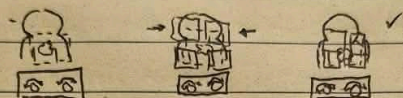
5. Collimation Selection: the patient will be zoomed for the injury and then the player must adjust the collimation knobs (to make it wider, smaller, longer, shorter)



- Right knob: adjusts the light on the ~~the~~ affected area horizontally. ex. from square to rectangle (landscape)

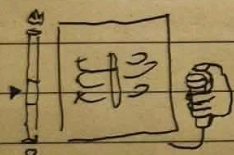
- Left knob: adjusts the light on the affected area vertically. ex. (portrait) $\updownarrow$

- so they will try to align the light with a preset marker size for the specific injury.



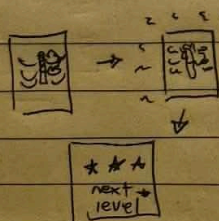
- marker is red and if fully aligned it will flash green then will move on to the next step which taking the xray image

6. The screen will look like an xray image and a hand on the side will press the button on the right time for the exposure. it will be a quick time event type of game.



- red: film is overexposed turns the film fully white
- green: just right sees the injury perfectly
- yellow: doest have enough exposure cant see much of the injury

- Once completed it will show the xray film in zoomed in and affects will come out for successfully completing the first level



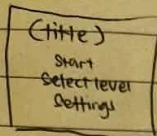
confetti will pop out as sign of ~~success~~ ^{success}

- star screen pops up and next level will appear and same process again but harder to adjust.

- | | |
|-------------------------------|-------------------------|
| level 1: Chest xray injuries | level 5: Skull injuries |
| level 2: upper limbs injuries | |
| level 3: lower limbs injuries | |
| level 4: abdominal injuries | |

Opens the App: sees title menu (includes start game, levels, & settings)

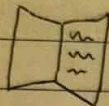
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Start game: level 1 example step by step:

1. patient enters and gives file to the radtech

2. radtech opens file and reads case injury ~~for the patient~~



in file - patient comes in for chest xray

- injury: broken left rib (due to riding accident)

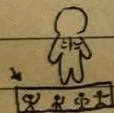
3. After reading file close the file and assess the patient.

- game will locate the injured area and by pressing on the injury it will zoom in to the injury (injury will be highlighted in red)



3. after assessing the patient they will now start positioning them into the proper positioning for the specific injury.

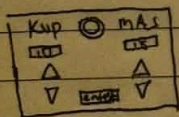
- They will select the appropriate pose in the following choases given in the bar selection below the screen



- if selected correct they will move onto the next step

- if selected incorrect the screen will flash red (every level will have specific positions per injury)

4. Next Step is to choose Radiation level and how long it should last (KVP, MAS)



- player must select the right amount of KVP & MAS per injury

- if chosen correctly the light on the machine turns green. So they start with KVP if green they move on to MAS if incorrect they must start again and light turns red

(instruction for chatbox: Select the right amount of KVP & MAS, the chest is sensitive so ~~don't~~ do not reach above 16 KVP and 3 secs MAS)

- KVP setting is 1 to 20 - MAS 1.5, 2.5 etc

- if light turns both green they may press the enter button to move onto the next step

